The Aether

Specification and Deployment Guide

SubThought Corporation
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The Aether: Specification and Deployment Guide

Email Premise.AI@Gmail.com for inquiries.

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Contents

1 Overview

The Aether is a Psyche: the interface between a GIL mind and the internet-as-world. It mediates two flows — percepts traveling inward from the world to the mind's mechanisms, and actuations traveling outward from the mind's mechanisms to the world. The Aether unifies the action surfaces of OmegaClaw (ASI Alliance / OpenCog Hyperon) and OpenClaw (the open-source agent framework) into a single coherent interface.

Where the NAO Psyche interfaces with a humanoid robot and the Ìtę Psyche interfaces with a game world, the Aether interfaces with the internet itself — web services, file systems, shell processes, messaging channels, browser automation, knowledge stores, external APIs, and physical devices via ROS 2.

All percepts from the Aether carry :Modality Aether. The :Channel slot selects the domain. The :Data slot names the parsable structure type. The :Content slot carries the idiom.

2 The Percept Relation

The Percept relation from totality.theory is canonical. It does not change for the Aether.

```
(relation Percept
  :M                ; lexified monad
  :Modality         ; which Psyche
  :Channel          ; sub-channel
  :Address  nil     ; URL or source
  :Data     nil     ; format type
  :Content  nil     ; idiom {:Slot value ...}
)
```

2.1 Slot Semantics

Slot	Meaning	Aether Examples
`:Modality`	Which Psyche delivered this percept.	Aether (always)
`:Channel`	Sub-channel within the Psyche.	Web, FileSystem, Process, Message, Browser, Memory, Service, System, Embodiment
`:Address`	URL or source of the data.	"https://...", "file:///frame.png", nil
`:Data`	Data format or parsable structure type.	search-result, page-content, process-output, media, message-received, knowledge-result
`:Content`	Idiom with the actual structured data.	{:Query "..." :Rank 1 :Title "..."}
`:Moment`	When the percept was received.	\@m{2026060100000000}

3 Percepts by Channel

All percepts from the Aether have :Modality Aether. The :Channel identifies the domain, :Data the format, :Address the source, and :Content the structured data idiom.

3.1 Channel Web

:Data	:Address	:Content (idiom)	Source
`search-result`	"https://..."	{:Query ... :Rank 1 :Title ... :Snippet ...}	Both
`page-content`	"https://..."	{:Title ... :ContentType "text/html" :Size 48000}	Both
`api-response`	"https://api..."	{:Method GET :Status 200 :Body {...} :Latency 120}	Both

3.2 Channel FileSystem

:Data	:Address	:Content (idiom)	Source
`file-content`	"file:///home/x"	{:ContentType "text/plain" :Size 2048 :Modified ...}	Both
`directory-listing`	"file:///home"	{:Entries {{:Name "src" :Type directory} ...}}	Both
`file-event`	"file:///home/x"	{:EventType modified :Size 2100}	OpenClaw

3.3 Channel Process

:Data	:Address	:Content (idiom)	Source
`process-output`	nil	{:Command "ls -la" :Stdout ... :ExitCode 0 :Duration 45}	Both
`background-output`	nil	{:SessionId "bg-42" :Stdout ... :Stderr ""}	OpenClaw

3.4 Channel Message

:Data	:Address	:Content (idiom)	Source
`message-received`	"telegram://room-7"	{:Sender "alice" :SenderType human :Text "hello"}	Both
`agent-reply`	"agentverse://agent-3"	{:Protocol agentverse :SessionId "s9" :Body {...}}	Both
`webhook-event`	"https://hooks..."	{:Source github :EventType push :Body {...}}	OpenClaw

3.5 Channel Browser

:Data	:Address	:Content (idiom)	Source
`browser-content`	"https://..."	{:Title ... :Selector "article" :ContentType text}	OpenClaw/Port42
`media`	"file:///frame.png"	{:Url ... :Width 1280 :Height 800 :Format png}	OpenClaw/Port42
`accessibility-tree`	"https://..."	{:NodeCount 340 :Tree {...}}	OpenClaw

3.6 Channel Memory

:Data	:Address	:Content (idiom)	Source
`memory-result`	nil	{:Query ... :StoreType embedding :Results {{:Score 0.87} ...}}	Both
`knowledge-result`	nil	{:Query ... :Results {{:Reifier 12345 :Confidence 0.9} ...} :InferenceTrail {...}}	OmegaClaw

3.7 Channels Service, System, Embodiment

Service: service-resource, financial-data. System: schedule-event, connection-status. Embodiment: media, sensor-reading, motor-state, action-result, transcription, clipboard. See the Aether.md companion file for complete idiom definitions.

4 Common Content Idiom Slots

Slot	Type	Count	Notes
`:ContentType / :Format`	literal	22	Describes the kind of data
`:Size / :Width / :Height / :Duration`	integer	15	Dimensional measure
`:Source / :Service / :Sender / :DeviceId`	literal	16	Who or what produced this data
`:Title`	string	5	Human-readable label
`:SessionId / :ThreadId / :JobId`	string	5	Session context
`:Status / :ExitCode`	literal/int	4	Outcome indicator
`:Confidence / :Score`	real	3	Certainty measure
`:InferenceTrail`	list	1	Reasoning provenance (OmegaClaw)

5 Actuations

Actuations are dispatched as ATTEMPT structures. The :Action slot names the literal. The :Parameters slot carries an idiom.

```
[ATTEMPT :Action exec  :Parameters {:Command "ls -la" :Timeout 30000}  :Token ?tok  :By \@m{...}]
```

The Aether defines 43 actuation literals across 10 domains.

Domain	Count	Literals
Shell Execution	4	exec, exec-background, process-kill, process-write
File System	6	file-create, file-overwrite, file-edit, file-patch, file-delete, file-move
Web Interaction	7	web-search, web-fetch, browser-navigate, browser-click, browser-type, browser-execute, browser-capture
Communication	5	message-send, agent-send, agent-spawn, notify, email-send
Memory	6	memory-store, memory-delete, knowledge-assert, knowledge-retract, nal-infer, pln-infer
Content Generation	3	generate-text, generate-image, render-ui
Automation	2	cron-schedule, cron-cancel
External Services	1	rest-call
Device / Embodiment	7	drive, gimbal, joint, light, speak, clipboard-write, automation-run
Capture	2	capture-screen, capture-camera

6 Cross-Cutting Structure

6.1 The Four Tuple Types

```
PERCEPT:  [PERCEPT :Modality Aether :Channel C :Address A :Data D :Content {:...}
:Moment M :Token T]
ATTEMPT:  [ATTEMPT :Action A :Parameters {:...} :Token T :By M]
RESULT:   [RESULT  :Action A :Status S :Reason R :Moment M :Token T]
URGE:     [URGE    :Need N :Source S :Delta D :Moment M :Token T]
```

The mind learns what its body did through perception. The result of an ATTEMPT comes back as a RESULT sent to the Perceiver — not the Executor. This is proprioception.

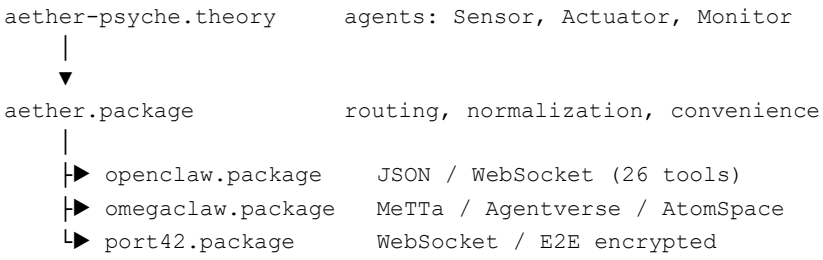
6.2 Actuation → Percept Pairings

:Action	Produces :Data
`exec, exec-background`	process-output
`file-create through file-patch`	file-event
`web-search`	search-result
`web-fetch`	page-content
`browser-navigate, click, type`	browser-content or media
`browser-capture`	media
`message-send, agent-send`	message-received or agent-reply
`memory-store`	memory-result
`knowledge-assert`	knowledge-result
`nal-infer, pln-infer`	knowledge-result (with :InferenceTrail)
`rest-call`	api-response
`capture-screen, capture-camera`	media
`cron-schedule`	schedule-event (future)
`drive-command, joint-command`	action-result, motor-state

7 Architecture

The Aether mediates between the mind (Perceiver/Executor) and three adapter frameworks (OpenClaw, OmegaClaw, Port42). The theory's agents call the aether.package, which routes to the correct adapter package, which translates to tell/ask calls over TCP, HTTPS, or pipes.

7.1 Package Layering



8 Deployment File Set

File	Category	Description
openclaw.package	Adapter	OpenClaw protocol: 30+ functions, generic dispatch
omegaclaw.package	Adapter	OmegaClaw protocol: 20+ functions, NAL/PLN, AtomSpace
port42.package	Adapter	Port42 protocol: 20+ functions, browser, audio, camera
aether.package	Unified Layer	Routing, normalization, convenience functions
aether-psyche.theory	Theory	Three agents: Sensor, Actuator, Monitor
Aether.daicho	Configuration	Adapter URLs, routing, polling, urge thresholds
aether.suites	Tests	9 suites: lifecycle through full end-to-end

9 Adapter Packages

openclaw.package wraps OpenClaw's tool groups (runtime, fs, web, ui, sessions, memory, messaging, automation, clawbody) into 30+ functions prefixed oc-, plus a generic oc-dispatch.

omegaclaw.package wraps OmegaClaw's skill categories (shell, agentverse, atomspace, reasoning, channels, memory) into 20+ functions prefixed omega-, plus omega-dispatch.

port42.package wraps Port42's bridge API domains (browser, screen, camera, audio, clipboard, automation, notify, rest) into 20+ functions prefixed p42-, plus p42-dispatch.

10 The Theory

aether-psyche.theory defines three agents. Sensor polls channels at configured intervals and wraps raw data as PERCEPT tuples. Actuator listens for ATTEMPT tuples from the Executor and dispatches through aether-act. Monitor checks homeostatic conditions (connectivity, inbox backlog, disk usage) and emits URGE tuples when thresholds are crossed.

On startup, the theory registers the Aether device with the Registrar, connects all adapters marked :AutoConnect yes, and starts the three agents.

11 Configuration

Section	Contents
`[Psyche]`	Agent URL, port, delay, modality literal
`[OpenClaw]`	URL, auto-connect, timeout, channels, capabilities
`[OmegaClaw]`	URL, auto-connect, timeout, channels, capabilities
`[Port42]`	URL, auto-connect, timeout, channels, capabilities
`[Routing]`	Per-channel adapter priority lists
`[Polling]`	Per-channel polling intervals in Moments
`[Urges]`	Homeostatic need definitions with thresholds

12 Framework Asymmetries

OmegaClaw has, OpenClaw lacks:

Formal symbolic reasoning (NAL, PLN) with auditable proof trails. Self-modifying control loop (MeTTa runtime introspection). Three-tier memory with typed hypergraph knowledge store (AtomSpace). Continuous autonomous execution loop.

OpenClaw has, OmegaClaw lacks:

Full browser automation (Playwright-based, multi-profile). 25+ messaging channel integrations. 5,400+ community skills via ClawHub and MCP. Structured file editing tools. Background process management with PTY. Cron-based scheduling. ROS 2 embodiment bridge (ClawBody).

13 Domain Coverage

Domain	:Channel	Percept :Data	Actuations	Source
`Local Compute`	FileSystem, Process	file-content, process-output, etc.	exec through file-move	Both
`Web`	Web	search-result, page-content, api-response	web-search, web-fetch, rest-call	Both
`Communication`	Message	message-received, agent-reply, webhook	message-send through email-send	Both
`Memory`	Memory	memory-result, knowledge-result	memory-store through pln-infer	Both
`Browser`	Browser	browser-content, media, a11y-tree	browser-navigate through capture	OC/P42
`External`	Service	service-resource, financial-data	rest-call	Both
`Embodiment`	Embodiment	media, sensor, motor, action, etc.	drive through capture-camera	OC/P42
`Automation`	System	schedule-event, connection-status	cron-schedule, cron-cancel	OpenClaw

14 Etymology

Aether (Greek Αἰθήρ, Aithēr) — from the verb αἴθω (aithō), “to burn, to ignite.” Originally the bright upper atmosphere that the gods breathed. Aristotle elevated it to the fifth element — the quinta essentia. In modern physics, the luminiferous aether was the medium through which electromagnetic waves propagated. Every era of meaning reinforces the same core idea: Aether is the medium through which things propagate. The internet is exactly this.

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